

ABDUL GAFFOOR MOHAMED SAMHI

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PROFESSIONAL SUMMARY

Third-year BSc (Hons) Computer Engineering undergraduate at the University of Ruhuna with hands-on experience in Java Spring Boot backend development, full-stack web applications, machine learning validation, DevOps (Docker, CI/CD) and TCP network programming. Delivered five projects including an independently built full-stack healthcare platform and led a four-member machine learning team. Seeking an internship in Software Engineering, AI/ML, DevOps or Networking.

EDUCATION

BSc (Hons) Computer Engineering | University of Ruhuna, Sri Lanka

December 2024 - Expected December 2028 | Currently Third Year, Semester 5

Relevant coursework: Data Structures & Algorithms, Advanced Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Machine Learning, Operating Systems & Network Programming, DevOps Engineering, Software Engineering

TECHNICAL SKILLS

Programming Languages: Java, Python, C, C++, C#, JavaScript, TypeScript

Backend & Web: Spring Boot, Spring Security, REST APIs, Node.js, Express.js, Next.js, React, JavaFX

Databases: PostgreSQL, MySQL, SQL, relational database design

Machine Learning: Pandas, NumPy, Scikit-learn, Matplotlib, regression, GroupKFold, model evaluation (MAE, RMSE, R2)

DevOps & Tools: Git, GitHub, Docker, Docker Compose, GitHub Actions (CI/CD), Maven, Jupyter Notebook, VS Code

Networking & Security: TCP/IP, Java Socket programming, multithreading, JWT, role-based access control, BCrypt, Wireshark

PROJECTS

LankaCare - Hospital & Healthcare Management System | Sole Full-Stack Engineer

Technologies: Java, Spring Boot, Spring Security, JWT, REST APIs, Next.js, React, TypeScript, Git

- Designed the full-stack architecture and built the Spring Boot backend and REST APIs for appointments, health records, prescriptions and hospital workflows.
- Implemented JWT-based authentication and role-based access control, and integrated the Next.js frontend with backend services and database persistence.

Smart Blood Glucose Prediction Using Machine Learning | Project Lead (4-member team)

Technologies: Python, Pandas, NumPy, Scikit-learn, Matplotlib

- Led the team in forecasting blood glucose 30 minutes ahead on public CGM datasets, comparing Ridge, KNN, SVR, Random Forest and Gradient Boosting against a persistence baseline.
- Designed participant-wise GroupKFold validation and fold-only preprocessing to prevent data leakage; evaluated with MAE, RMSE and R2 at participant level.

Emergency Resource Coordination System (ERCS) | DevOps & Cloud Infrastructure Engineer (2-member team)

Technologies: Docker, Docker Compose, GitHub Actions, Node.js, Express.js, React, PostgreSQL

- Containerized frontend and backend services with Docker and Docker Compose for reproducible development and deployment.
- Built GitHub Actions CI/CD workflows for install, test and build, and managed environment-based configuration and secrets.

Gramalink LK - Real-Time Coordination Network | Network & Concurrency Engineer (3-member team)

Technologies: Java 21, JavaFX, Socket/ServerSocket, TCP/IP, Jackson, PostgreSQL, Wireshark

- Built a multi-client TCP server using ExecutorService threads, with a custom length-prefixed JSON protocol and authenticated sessions.
- Implemented heartbeat monitoring, reconnection handling and thread-safe shared state; debugged traffic using Wireshark.

Gridora - Digital Twin Platform for Power Plants | Backend Systems Engineer (3-member team)

Technologies: Backend services, database, real-time data integration (github.com/Gridora-Systems)

- Contributing to backend architecture for collecting, processing and storing plant telemetry and synchronizing it with the Digital Twin model.
- Supporting service design for simulation, what-if analysis and a scalable multi-plant architecture.

LEADERSHIP, MEMBERSHIP & ACTIVITIES

- Project Lead, Smart Blood Glucose Prediction (4-member team)
- Student Member, Institution of Engineers Sri Lanka (IESL), Membership No. S-3388
- Participant, IEEEExtreme Programming Competition and RedCypher Competition

LANGUAGES

Tamil (Native) | English (Working Proficiency)